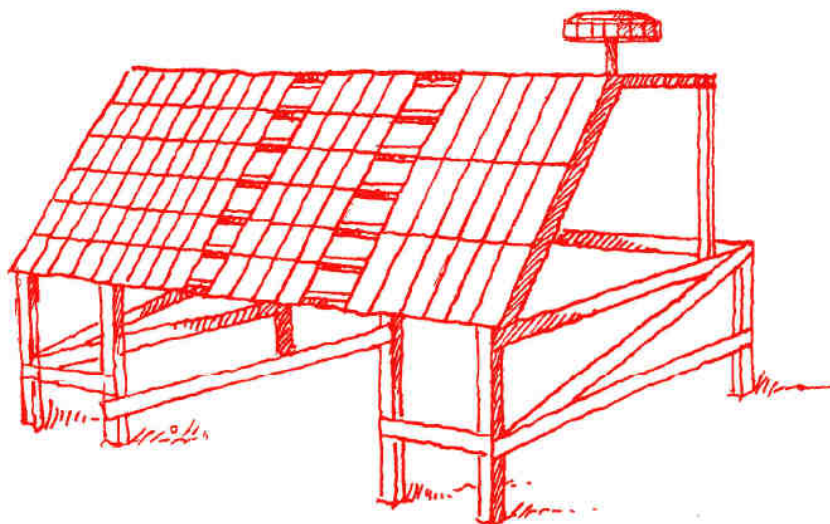


CONVENTION ON LONG-RANGE TRANSBOUNDARY AIR POLLUTION

UN/ECE INTERNATIONAL CO-OPERATIVE PROGRAMME
ON EFFECTS ON MATERIALS, INCLUDING HISTORIC
AND CULTURAL MONUMENTS



Report No. 53
Results from the multi-pollutant programme
Corrosion attack on carbon steel after 1year of exposure
(2005-06)

March 2008

PREPARED BY THE SUB-CENTRE FOR STRUCTURE METALS AND TRENDS



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Prague/ the Czech Republic

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after 1 year of exposure (2005-06)

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Summary

This report details the experimental method and gives results of evaluation of carbon steel corrosion after exposure in unsheltered position at 30 test sites for 1 year in period 2005 - 06 in frame of the UN ECE ICP Effects on Materials. The report summarises the results of evaluation of corrosion gains, corrosion losses and evaluation of selected corrosion active water soluble compounds in rust layers.

The tolerable level for carbon corrosion loss (157 g.m^{-2}) is overcome on 4 test sites. During this exposure the corrosion loss of carbon steel increased in comparison to previous period significantly on test site No 23 (Birkenes) and partly on test sites No 21 and 24 (Figure 11). In context to corrosion loss of zinc and limestone which show out this phenomenon more significantly for these test sites.

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Introduction

The UN ECE International Co-operative Programme was performed to a quantitative evaluation of the effect of sulphur pollutants in combination with NO_x and other pollutants as well as climatic parameters on the atmospheric corrosion of important materials. The programme was based on an international field exposure on 39 test-sites in a wide geographical zone of the 13 Signatories of the Convention on Long-Range Transboundary Air Pollution. The basic field exposure was performed in 1987-1995.

Within 1985 - 1995 the environmental situation have been changed. The SO₂ concentrations significantly decreased in many European countries and the pollution situation changed to multipollutant. This phase of exposure programme (1997-2001) is focused on the possible synergetic effects of NO_x, SO₂ and O₃ on a range of materials including carbon steel, zinc, copper, bronze, painted steel, glass and calcareous stone. The following list names the responsible sub-centres of this programme:

- Swedish Corrosion Institute, Stockholm, Sweden (main centre),
- ENEA, Rome, Italy,
- SVUOM Ltd. , Prague, the Czech Republic,
- Swiss Federal Laboratories for Material Testing and Research, Dübendorf, Switzerland,
- Norwegian Institute for Air Research, Kjeller, Norway,
- Building Research Establishment, Garston Watford, the United Kingdom.

SVUOM has been sub-centre responsible for exposure and evaluation of samples of carbon steel in the frame of programme. The corrosion of unalloyed carbon steel is influenced to an high extent by environmental conditions mainly by acidifying airborne pollutants.

1. Methods

1.1. Characterisation of samples

Plates of unalloyed carbon steel (with C < 0.2 %, P < 0.07 %, Cr < 0.07 % according to CSN 11373) with dimensions 100 x 150 x 0,5 mm were used, triplicate per exposure period.

Before the exposure the samples were degreased in alkaline degreaser, rinsed with ethanol, dried and weighed.

1.2. Exposure of samples

Methods prescribed in the Report No 51 - Technical manual to the UN ECE International Co-operative Programme on Effects on Materials including Historic and Cultural Monuments [1] were applied for exposure of samples on test sites. Samples were exposed on racks in open atmosphere (unsheltered exposure) according to ISO 8565. For each exposure there had been used triplicate samples.

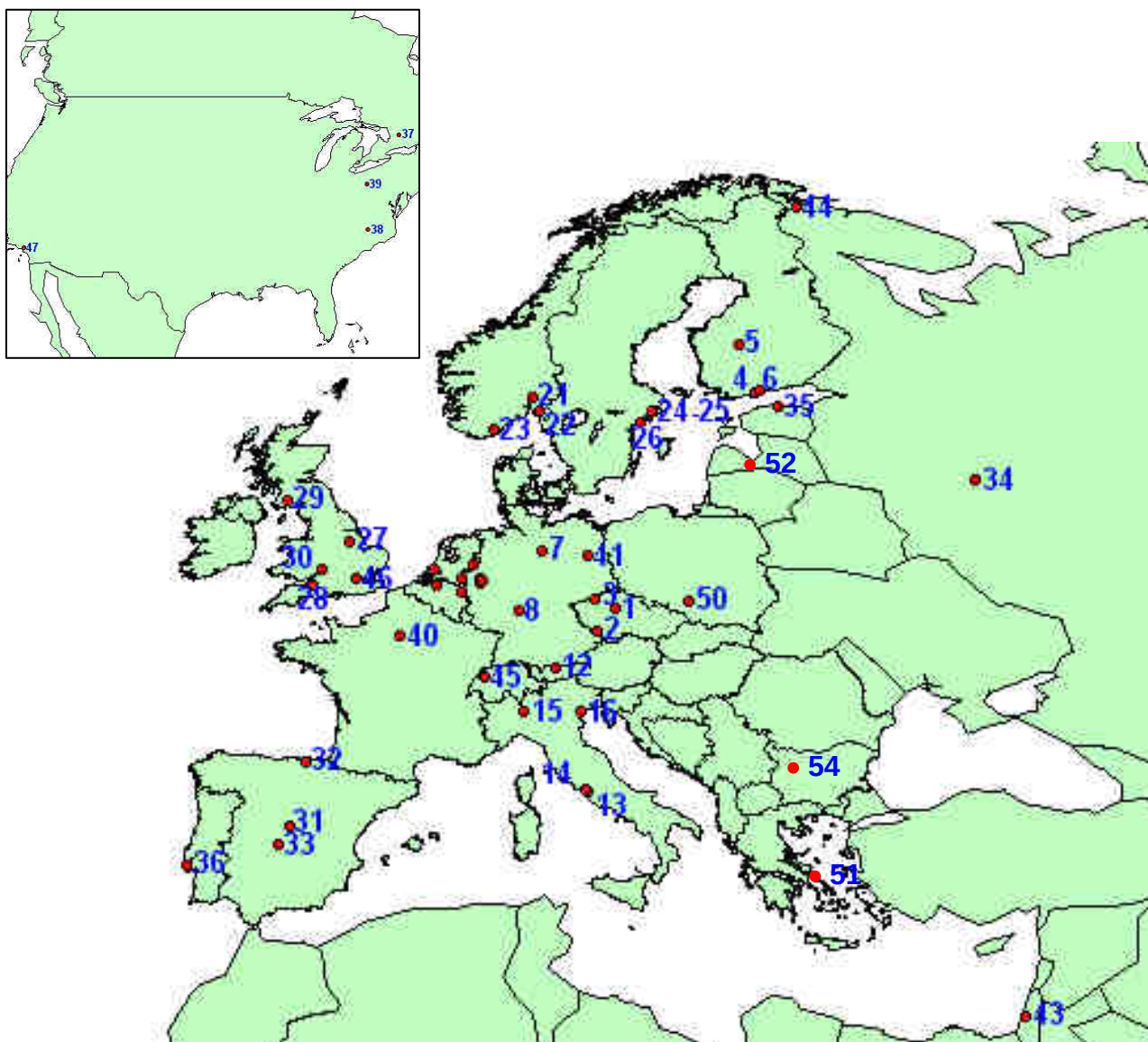
The list of test sites for the multi pollutant programme is in Table 1 and Figure 1. Exposure period has been 1 year (2005-06). The samples had been exposed on test site No 54 Sophia, Bulgaria in period 2006-2007.

Table 1 - The list of test sites

Country	Test site	Type of atmosphere
the Czech Republic	1 Prague 3 Kopisty	urban-industrial industrial
Germany	10 Bottrop 41 Berlin	industrial urban
Italy	13 Rome 14 Casaccia 15 Milan 16 Venice	urban rural urban-industrial urban
Norway	21 Oslo 23 Birkenes 44 Svanvik	urban rural rural
Sweden	24 Stockholm South 26 Aspvreten	urban rural
the United Kingdom	27 Lincoln Cathedral	urban
Spain	31 Madrid 33 Toledo	urban rural
Estonia	35 Lahemaa	rural
Canada	37 Dorset	rural
France	40 Paris	urban
Switzerland	45 Chaumont	rural
Poland	50 Katowice	industrial
Greece	51 Athens	urban
Latvia	52 Riga	urban
Bulgaria	54 Sophia	urban

The samples had not been exposed on test site No 27 from technical reason. The exposed samples had not been sent back to sub-centre from test sites No 10 and No 16.

Figure 1 – Map showing location of test sites



1.3. Measurement of environmental data

Basic climatic parameters (temperature, relative humidity and intensity of sunshine radiation), concentration of gaseous pollutants (SO_2 , NO_x , O_3) and precipitation (amount, pH, conductivity, amount of SO_4^{2-} , NO_3^- , Cl^- , and NH_4^+ , Na^+ , K^+ , Ca^{2+} , Mg^{2+} if available) were measured on each test site. All data were reported to, and completed by the Norwegian Institute for Air Research - NILU, which also checked the quality of the data [2].

The environmental data collected in the course of the multi-pollutant programme confirms the decreasing concentration of sulphur dioxide in atmosphere.

1.4. Evaluation of corrosion attack

The visual evaluation of layer of corrosion products was done to indicate general, non uniform corrosion process.

The basic information about corrosion behaviour of carbon steel in different environmental conditions gives the value of mass change and corrosion loss. After exposure samples were weighed (conditioned in laboratory conditions) and the corrosion products were removed using pickling method according to ISO 8407 (Annex A - solution C3.5.). In consecutive pickling cycles the corrosion layer was completely removed without dissolving the base metal. The repetition of the procedure stopped if the surface appears clean. The corrosion losses had been obtained by gravimetric method.

For selected samples of carbon steel the amount of water soluble salts in the corrosion products was performed by spectrophotometric analysis. There were analysed corrosion layers from samples exposed at test sites No 1, 3, 13, 15, 33, 40, 41, 50 and 51 (see 1.5).

1.5. Evaluation of corrosion active characteristics of rust layer

The analysis of corrosion active components and evaluation of some other characteristics of rust layers gives an important information on the interaction of environment with the steel surface and on expected function of rust in corrosion system including possible protective coatings. Analysis was performed on two rust sub-layers separate:

- the non-adherent rust sub-layer had been got by mechanical removal by fine steel brush separately for upper and ground side of exposed samples. The content of active components had been analysed after extraction by deionised water for 10 min. in ultrasonic wave bath.
- the adherent rust layer had been extracted by deionised water for 24 hs.

2. Results

2.1. Visual evaluation

An illustrative evaluation was done in combination with a digital photo of each sample. The layers of corrosion products of carbon steel had formed a typical structure, colour and thickness according exposure conditions - Figure 2.

Figure 2 – Carbon steel surface with corrosion products

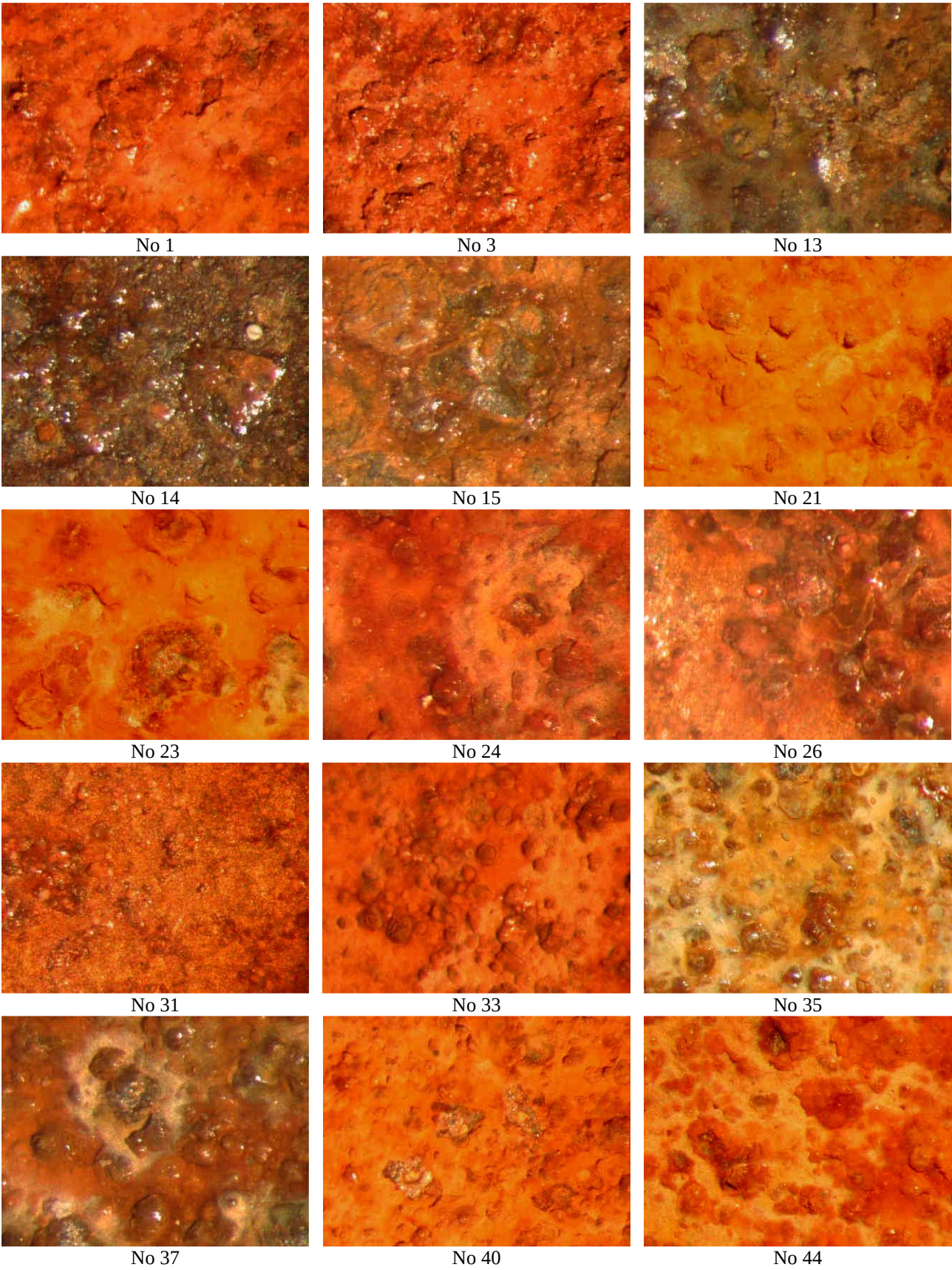
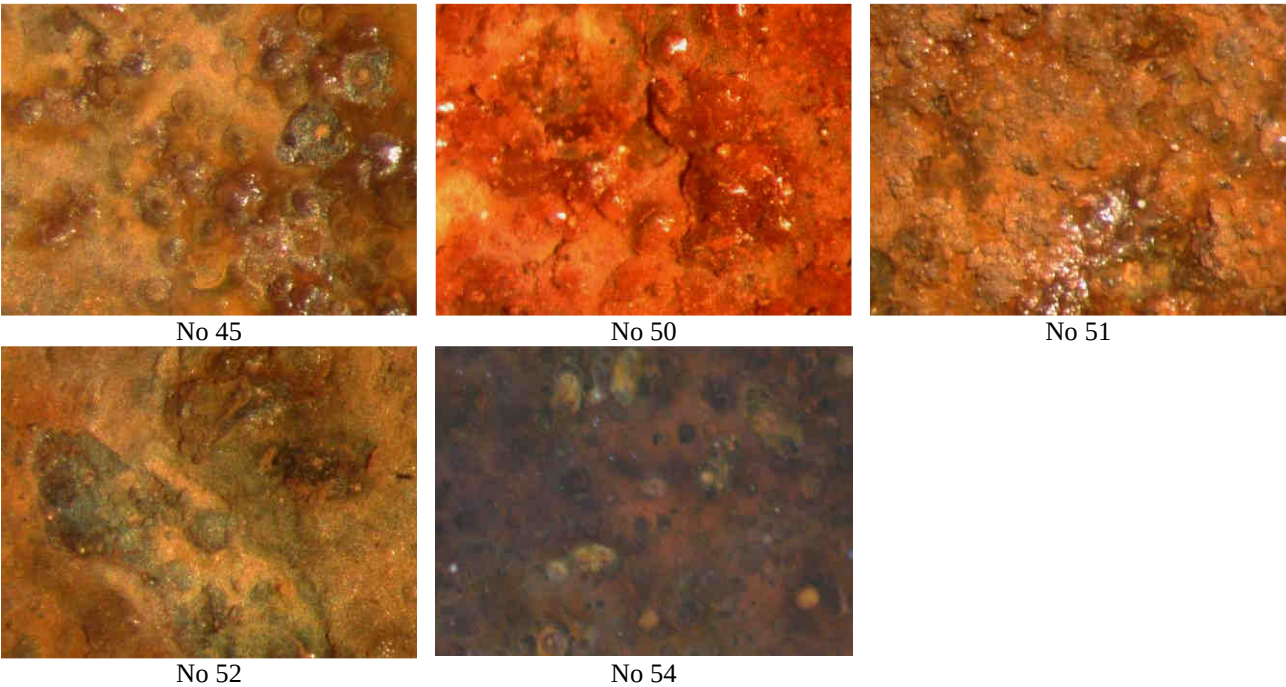


Figure 2 – Carbon steel surface with corrosion products - continue



2.2. Mass changes of corroded carbon steel

Mass changes of exposed samples represent differences of mass of corrosion products originated at impact of environment and part of them removed by run-off erosion and fall out of non adherent particles and sub-layers. Mass increase of exposed steel samples gives a complex and not well-defined information. Mass changes of exposed samples of carbon steel exposed at open atmosphere are summarised in Table 2 and Figure 3.

Figure 3 – Mass changes of carbon steel samples after 1 year exposure

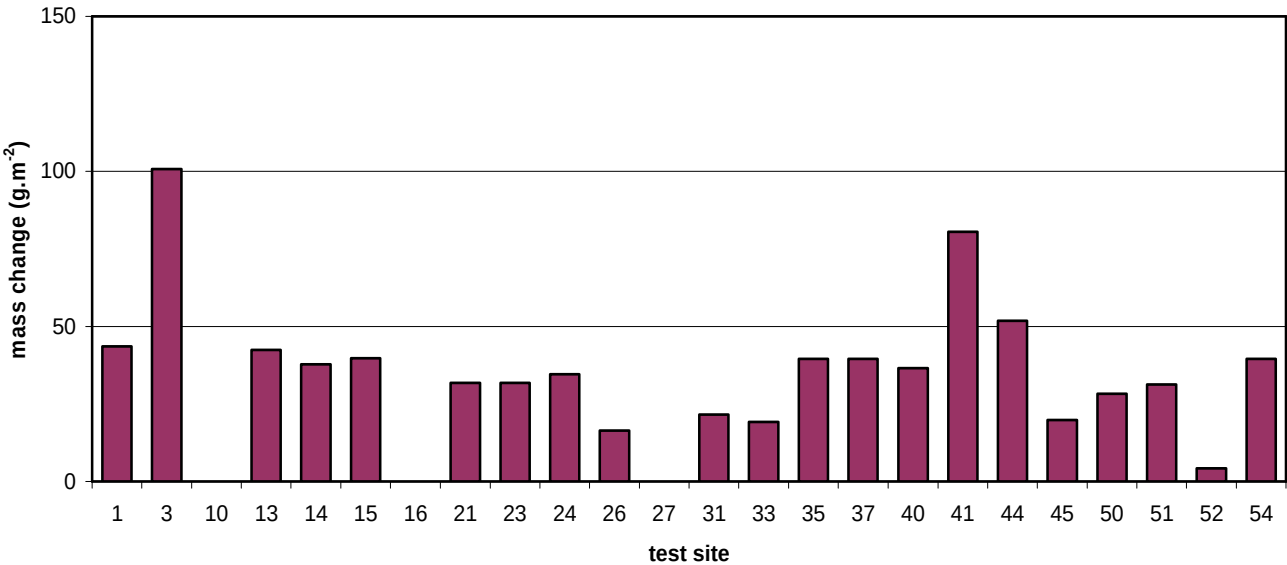


Table 2 - Mass change of carbon steel after 1 year exposure at open atmosphere

Test site	MC ₈₁ (g.m ⁻²)	MC ₈₂ (g.m ⁻²)	MC ₈₃ (g.m ⁻²)	MC (g.m ⁻²)
1 Prague	43,1067	44,2000	43,3367	43,5478
3 Kopisty	101,0033	100,9333	100,5500	100,8289
10 Bottrop	-	-	-	-
13 Rome	43,5600	40,9167	42,9500	42,4756
14 Casaccia	35,7500	38,0067	39,6133	37,7900
15 Milan	39,7367	38,2600	41,0767	39,6911
16 Venice	-	-	-	-
21 Oslo	30,3400	32,3233	32,5500	31,7378
23 Birkenes	32,3333	32,6533	30,4567	31,8144
24 Stockholm South	33,8500	34,2433	35,6867	34,5933
26 Aspvtren	16,1367	16,5400	16,5533	16,4100
27 Lincoln Cathedral	-	-	-	-
31 Madrid	19,7667	18,7533	26,2200	21,5800
33 Toledo	20,1633	18,8767	18,6733	19,2378
35 Lahemaa	40,3900	39,5933	38,5500	39,5111
37 Dorset	39,7433	39,6333	39,1500	39,5089
40 Paris	33,5133	38,1833	38,0467	36,5811
41 Berlin	83,0300	84,0467	74,4900	80,5222
44 Svanvik	50,9300	52,7433	51,7533	51,8089
45 Chaumont	19,0467	21,1733	19,2967	19,8389
50 Katowice	30,8633	27,9333	26,0567	28,2844
51 Athens	28,0467	33,8133	32,2500	31,3700
52 Riga	11,0067	1,9200	-0,2800	4,2156
54 Sohia	42,2233	38,9533	37,3767	39,5178

Relative part of non-adherent and adherent rust sub-layers on steel samples is presented in Table 3.

Table 3 – Comparison of non-adherent and adherent parts of rust layer

test site	non-adherent part* (g.m ⁻²)	adherent part** (g.m ⁻²)	non-adherent part (%)	adherent part (%)
1	46	99	32	68
3	71	213	25	75
13	41	106	28	72
15	53	104	34	66
33	27	38	42	58
40	38	85	30	70
41	168	142	54	46
50	57	197	22	78
51	46	103	31	69

Note: * The part removed from samples after exposure by fine steel brush.

** The difference between total corrosion mass loss and weight of non-adherent part of corrosion products.

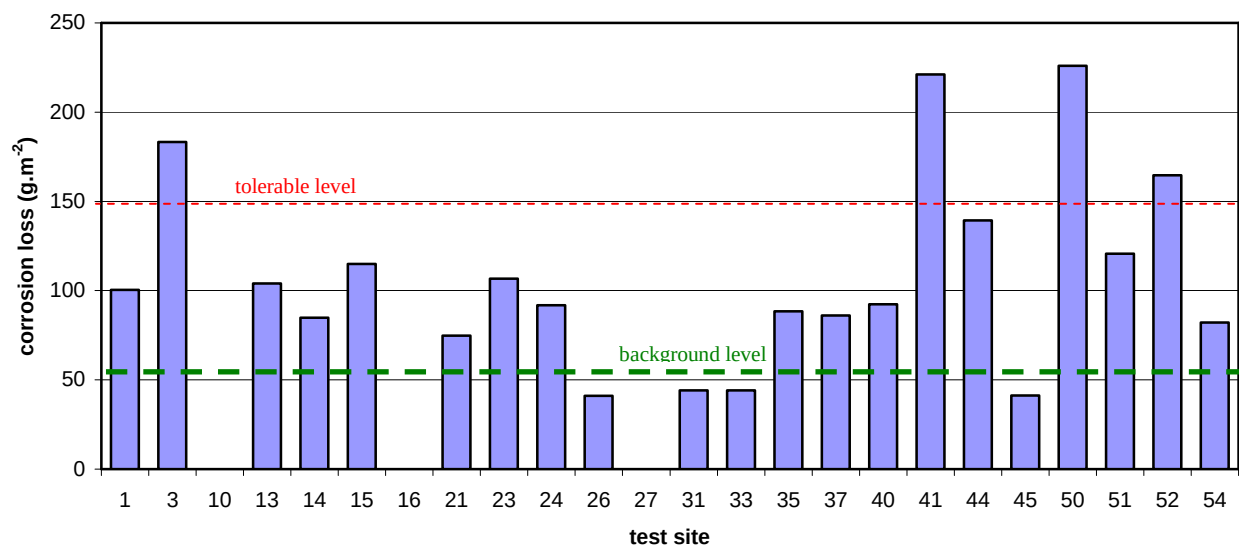
2.3. Corrosion losses of carbon steel

Corrosion losses for individual samples of carbon steel exposed in open atmosphere for 1 year exposure (2005/06) are summarised in Table 4 and presented in Figure 4. The test sites with the greatest corrosion losses of samples were No 41 (Berlin) and No 50 (Katowice). In Figure 4 the tolerable level for carbon corrosion loss (157 g.m^{-2}) is drawn – 4 test sites are up this level.

Table 4 - Corrosion losses (g/m^2) of carbon steel after 1 year exposure in open atmosphere

Test site	ML ₈₁ (g.m^{-2})	ML ₈₂ (g.m^{-2})	ML ₈₃ (g.m^{-2})	ML (g.m^{-2})
1 Prague	102,3449	101,7307	97,0443	100,3733
3 Kopisty	184,5789	180,5127	184,8000	183,2923
10 Bottrop	-	-	-	-
13 Rome	102,2400	100,0014	105,7959	102,6791
14 Cassacia	80,9085	84,9738	88,8587	84,9137
15 Milan	115,4080	115,1500	114,1483	114,9021
16 Venice	-	-	-	-
21 Oslo	72,6933	75,4067	76,1645	74,7548
23 Birkenes	104,8730	106,1536	108,9793	106,6686
24 Stockholm South	91,4009	91,1709	92,6800	91,7506
26 Aspvreten	39,6944	41,5089	42,2078	41,1370
27 Lincoln Cathedral	-	-	-	-
31 Madrid	40,1711	38,6189	53,5722	44,1207
33 Toledo	50,3663	40,3789	41,3561	44,0338
35 Lahemaa	90,8162	89,4160	85,0246	88,4189
37 Dorset	87,8989	84,8550	85,7263	86,1601
40 Paris	89,7217	93,0467	94,4414	92,4033
41 Berlin	227,4728	224,6739	211,4060	221,1842
44 Svanvik	153,2000	132,5739	132,1161	139,2967
45 Chaumont	39,9856	43,3189	40,2700	41,1915
50 Katowice	229,1836	226,1559	222,6286	225,9894
51 Athens	134,7148	112,9815	114,3519	120,6827
52 Riga	157,7093	166,9096	169,1910	164,6024
54 Sophia	82,0043	80,3034	84,0360	82,1145

Figure 4 – Corrosion loss of carbon steel after 1 year exposure



2.4. Analysis of corrosion products

Analysis of corrosion active compounds of rust layer is significant information of interaction between environment and corroded material. The detailed analysis had been performed for selected test sites only and the results are in Tables 5 and 6. The evaluated corrosive agents (Cl^- , SO_4^{2-} , NO_3^-) are located in preference in non-adherent sublayer (SO_4^{2-} , NO_3^-) or in adherent sublayer (Cl^-).

Table 5 – The analysis of water extract of surface - non-adherent part of rust layer

Test site	pH	conductivity [$\mu\text{S}/\text{cm}$]	Ion concentration [mg/m^2]		
			Cl^-	SO_4^{2-}	NO_3^-
1	6,49	18	0,00	2,20	8,69
3	6,28	12	0,00	3,82	12,65
13	6,06	17	0,40	1,42	4,21
15	6,13	17	0,28	2,34	6,93
33	5,96	14	0,00	0,93	2,45
40	5,99	22	0,15	1,24	6,14
41	6,35	19	0,00	3,22	21,62
50	6,06	24	0,30	1,64	8,00
51	6,09	25	0,00	1,86	4,24

Table 6 – The analysis of water extract of surface - adherent part of rust layer

Test site	pH	conductivity [$\mu\text{S}/\text{cm}$]	Ion concentration [mg/m^2]		
			Cl^-	SO_4^{2-}	NO_3^-
1	6,12	33	2,93	0,07	0,33
3	6,06	53	2,67	0,03	1,07
13	6,00	65	5,20	0,08	0,33
15	5,89	66	4,27	0,07	0,33
33	6,02	48	5,00	0,05	0,33
40	5,85	68	4,27	0,11	4,27
41	5,71	64	7,13	0,08	1,27
50	5,44	83	5,13	0,12	0,40
51	5,54	80	5,27	0,12	0,33

The adherent layers of corrosion product contain higher amount of soluble salts (see conductivity value in Tables 5 and 6) and the dominante compound is chloride. The highest content of chloride had been found for samples from test site No 41 – Berlin which is located directly in the highway. The non-adherent, upper layers of corrosion product contain maily the sulphate and the nitrate. The highest content of both ions had been found for samples from test site No 41 – Berlin and from test site No 3 Kopisty which is influnced by industrial pollution.

3. Treatment of results

3.1. Environmental data

In Table 7 there are summarized environmental data for exposure period. The newly measured environmental factors are gaseous HNO_3 and particles.

There is correlation between carbon steel corrosion loss and individual air gaseous pollution (Figure 5). The only gaseous pollution which has the significant effect onto carbon steel corrosion is SO_2 .

Figure 5 – The correlation between carbon steel corrosion loss and individual gaseous air pollution

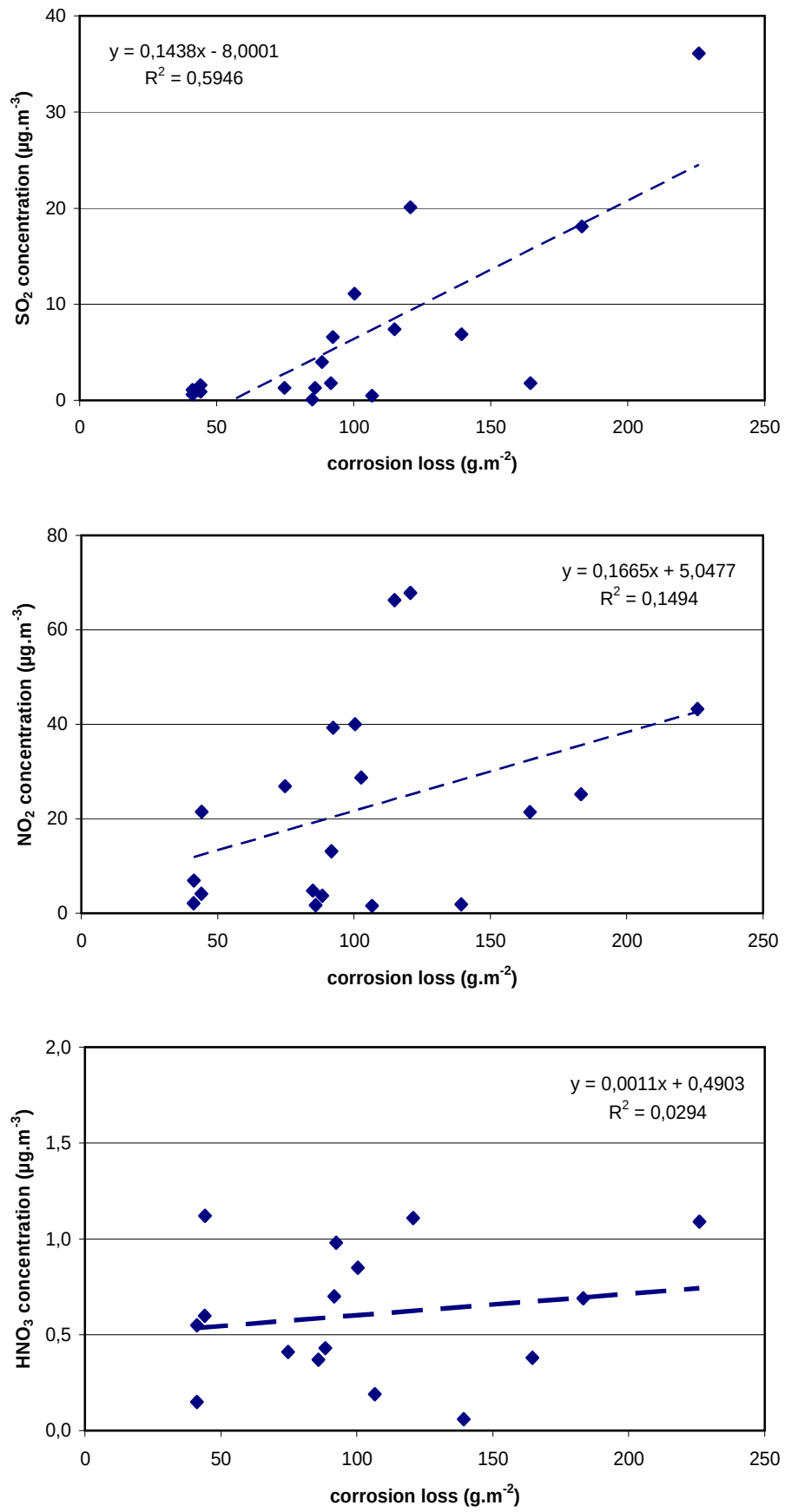


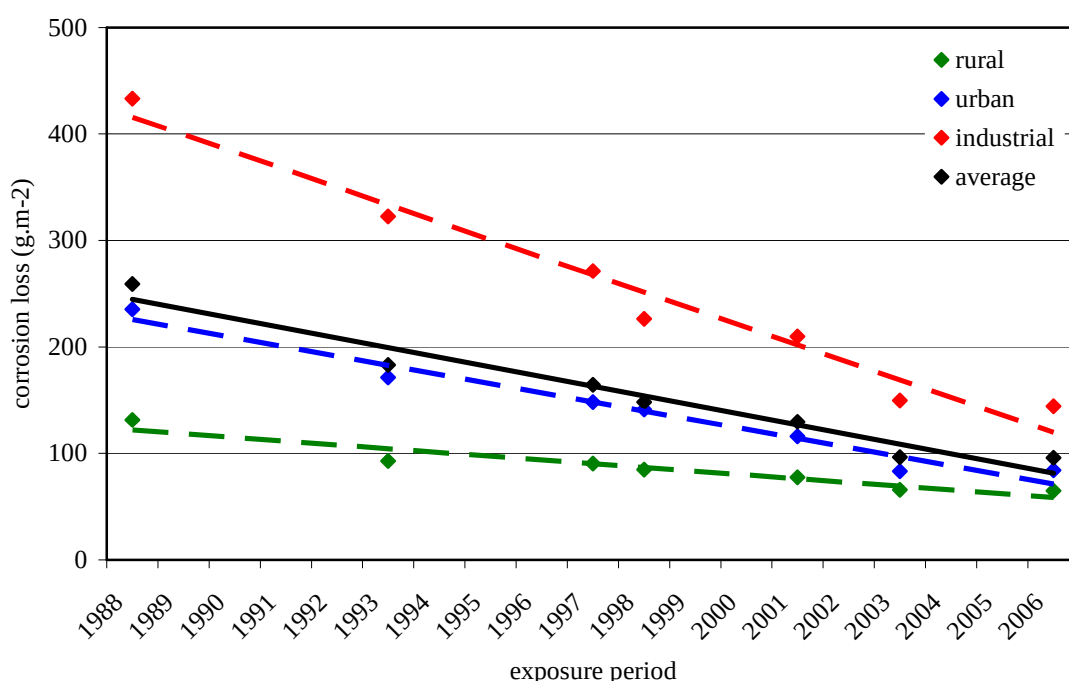
Table 7 - Average environmental parameters for exposure period 2005/06

Test site	CLIMATE			GASES				PRECIPITATION						PARTICLES			
	Temp °C	Rh %	Sun MJ/m ²	SO ₂ µg/m ³	NO ₂ µg/m ³	O ₃ µg/m ³	HNO ₃ µg/m ³	mm	pH	SO ₄ ²⁻ mg/l	NO ₃ ⁻ mg/l	Cl ⁻ mg/l	Cond µS/cm	PM ₁₀ µg/m ³	SO ₄ ²⁻ mg/m ² d	NO ₃ ⁻ mg/m ² d	Cl ⁻ mg/m ² d
1 Prague	9,3	74		11,1	40,0		0,85	491,4	5,7	6,9	8,3	2,2	55,4	20	1,01	0,98	0,28
3 Kopisty	8,3	76		18,1	25,2		0,68	442,2	5,9	8,4	5,6	1,2	43,8	21	0,91	0,60	0,09
10 Bottrop	11,0	77		16,4	33,2	37	0,74	761,3	5,8	10,7	8,5	5,9	35,7	20	1,22	0,57	0,69
13 Rome					28,7	19											
14 Casaccia	15,8	66		0,1	4,8	51											
15 Milan	14,0	56		7,4	66,3	28		632,0									
16 Venice	13,9	76		2,3				889,0									
21 Oslo	7,2	74		1,3	26,9	36	0,41	869,1						11	0,37	0,51	0,15
23 Birkenes	6,2	79		0,5	1,6	61	0,19	1623,1	4,6	0,4	0,5	1,6	20,9	5	0,21	0,36	0,32
24 Stockholm South	8,9	74		1,8	13,1	58	0,70							31	0,72	1,11	0,53
26 Aspvreten	6,3	84		0,6	2,1	44	0,15							4	0,11	0,19	0,05
27 Lincoln Cathedral																	
31 Madrid	15,3	56		1,6	21,5	49	1,12	399,0	6,3	0,4	0,3	0,4	12,4	15	0,39	0,98	0,14
33 Toledo	12,1	69		0,9	4,1	88	0,60	689,4	6,1	0,5	0,3	0,7	14,2	11	0,21	0,51	0,22
35 Lahemaa	5,2	80		3,9	3,6	62	0,43	402,8	4,2	0,3	0,3	0,4	22,6	11	0,27	0,39	0,17
37 Dorset				1,3	1,7	53	0,37							5	0,25	0,24	0,03
40 Paris	12,6	73		6,6	39,3	38	0,98	571,0	6,2	6,6	0,4	11,1		33	2,43	1,68	2,02
41 Berlin							0,53							95	1,64	1,09	2,64
44 Svanvik	0,3	78		6,9	1,9	67	0,19	711,4						9	0,24	0,17	0,19
45 Chaumont	6,2	80		1,1	6,9	86	0,55	1403,7	5,2	0,4	0,2	0,1	8,1	6	0,18	0,27	0,02
50 Katowice	7,5	76		36,1	43,2	38	1,09	673,8						29	1,34	0,82	0,50
51 Athens	18,3	70		20,1	67,8	20	1,11	374,2						146	14,20	3,37	15,97
52 Riga	7,0	76		1,8	21,4	16	0,38	545,3	6,4	0,6	0,4	1,6	39,3	57	2,01	1,03	2,52
54 Sophia																	

3.2. Comparison of 1 year corrosion losses in different exposure periods

In Table 8 and Figures 9 and 10 there is a comparison of corrosion losses after 1 year of exposure in different periods. There is evident that corrosion rate of carbon steel is decreasing during exposure time on all test sites. The only exception is corrosion loss of carbon steel in period 2005/06 at test sites No 21, 23 and 24. The increasing of corrosion loss at test site No 41 Berlin may be caused by specific conditions (the rack is located in the middle of the highway).

Figure 10 - Comparison of average values of corrosion loss from 1987 till 2006 (16 selected test sites only)



The decreasing trend in SO₂ pollution in regions where the test sites are located is now practically terminated and in some location the air pollution increased again – e.g. the Czech republic. The reason for this is suitable pricing policy of energy. The detailed evaluation can be done for test site Kopisty where the carbon steel and zinc samples had been exposed repeatedly and the SO₂ concentration increased in the last 2 years. The corrosion rate of carbon steel and zinc increased in this location again (Figure 10).

Figure 9 – Corrosion loss (g.m⁻²) after 1 year of exposure in various periods on selected test sites (since 1987/88 to 2005/06)

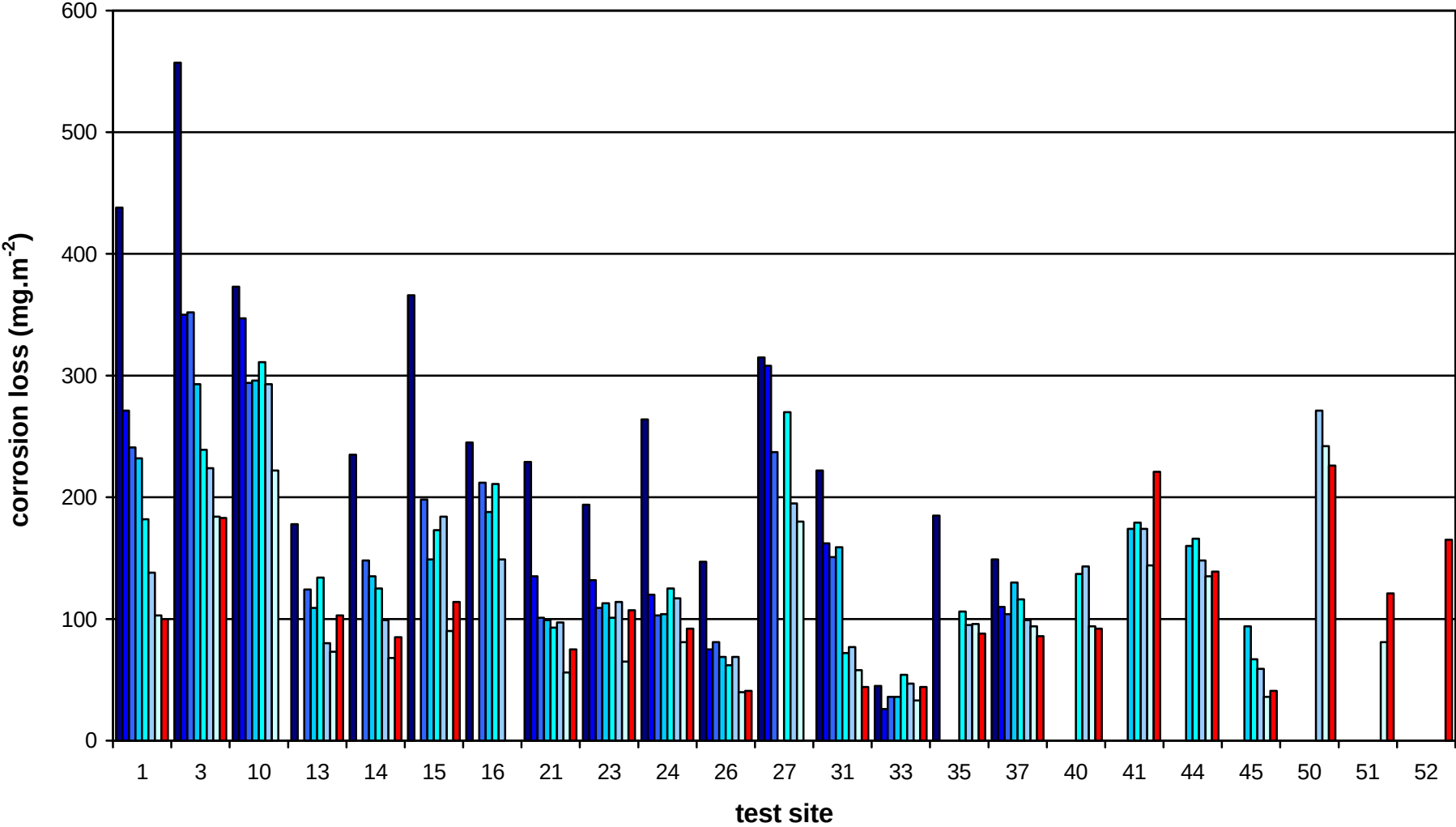


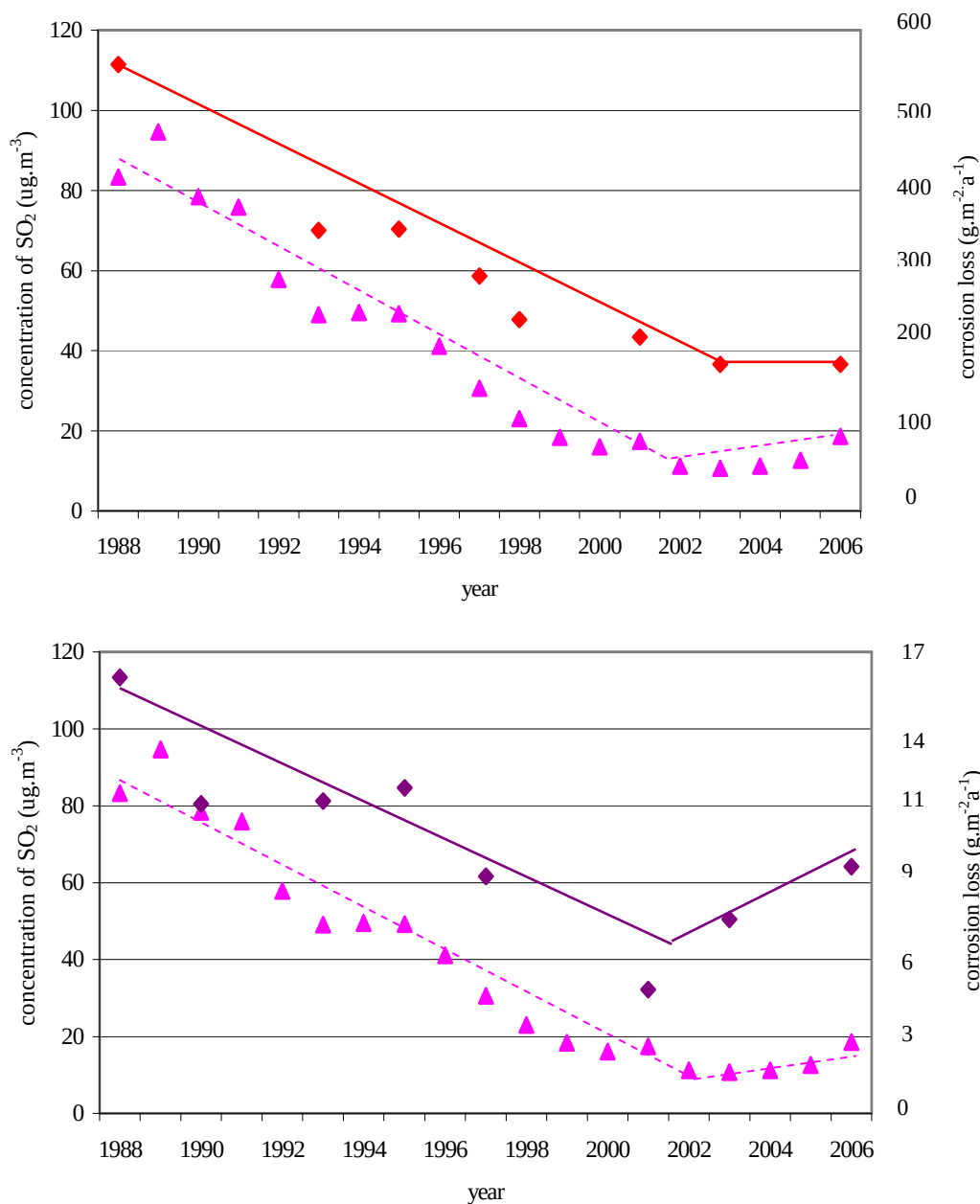
Table 8 - Comparison of corrosion losses (g/m^2) of carbon steel in different periods

Test site	period 1987-88	period 1992-93	period 1994-95	period 1996-97	period 1997-98	period 2000-01	period 2002-03 ¹	period 2005-06
1 Prague	438 C4	271 C3	241 C3	232 C3	182 C2	138 C2	103 C2	100 C2
3 Kopisty	557 C4	350 C3	352 C3	293 C3	239 C3	224 C3	184 C2	183 C2
10 Bottrop	373 C3	347 C3	294 C3	296 C3	311 C3	293 C3	222 C3	-
13 Rome	178 C2	-	124 C2	109 C2	134 C2	80 C2	73 C2	103 C2
14 Casaccia	235 C3	-	148 C2	135 C2	125 C2	98 C2	68 C2	85 C2
15 Milan	366 C3	-	198 C2	149 C2	173 C2	184 C2	90 C2	115 C2
16 Venice	245 C3	-	212 C3	188 C2	211 C3	149 C2	-	-
21 Oslo	229 C3	135 C2	101 C2	99 C2	93 C2	97 C2	56 C2	75 C2
23 Birkenes	194 C2	132 C2	109 C2	114 C2	101 C2	114 C2	65 C2	107 C2
24 Stockholm	264 C3	120 C2	103 C2	104 C2	125 C2	117 C2	81 C2	92 C2
26 Aspvreten	147 C2	75 C2	81 C2	69 C2	62 C2	69 C2	40 C2	41 C2
27 Lincoln Cathedral	315 C3	308 C3	237 C3	-	270 C3	195 C2	180 C2	-
31 Madrid	222 C3	162 C2	151 C2	159 C2	72 C2	77 C2	58 C2	44 C2
33 Toledo	45 C2	26 C2	36 C2	36 C2	54 C2	47 C2	33 C2	44 C2
35 Lahemaa	185 C2	-	-	-	106 C2	95 C2	96 C2	88 C2
37 Dorset	149 C2	110 C2	104 C2	120 C2	116 C2	99 C2	94 C2	86 C2
40 Paris	-	-	-	-	137 C2	143 C2	94 C2	92 C2
41 Berlin	-	-	-	174 C2	179 C2	174 C2	144 C2	221 C3
44 Svanvik	-	-	-	160 C2	166 C2	148 C2	135 C2	139 C2
45 Chaumont	-	-	-	94 C2	67 C2	58 C2	36 C2	41 C2
50 Katowice	-	-	-	-	-	271 C3	242 C3	226 C3
51 Athens	-	-	-	-	-	-	81 C2	121 C2
52 Riga	-	-	-	-	-	-	-	165 C2

C2, C3 , C4 corrosivity categories according ISO 9223

¹ exposure in frame of EU project MULTI-ASSESS

Figure 10 - Trend of decreasing of concentration of SO₂ and corrosion rate of carbon steel and zinc (test site No 3 Kopisty)



3.3. Evaluation of results – dose/response functions for carbon steel

On the base of previous exposure programs (MULTI-ASSESS – equation [1], ISO 9223 – equation [2]) there were derived various D/R functions for carbon steel:

$$ML = 29.1 + \{ 21.7 + 1.39[SO_2]^{0.6} Rh_{60} e^{f(T)} + 1.29 Rain[H^+] + 0.593 PM_{10} \} t^{0.6} \quad [1]$$

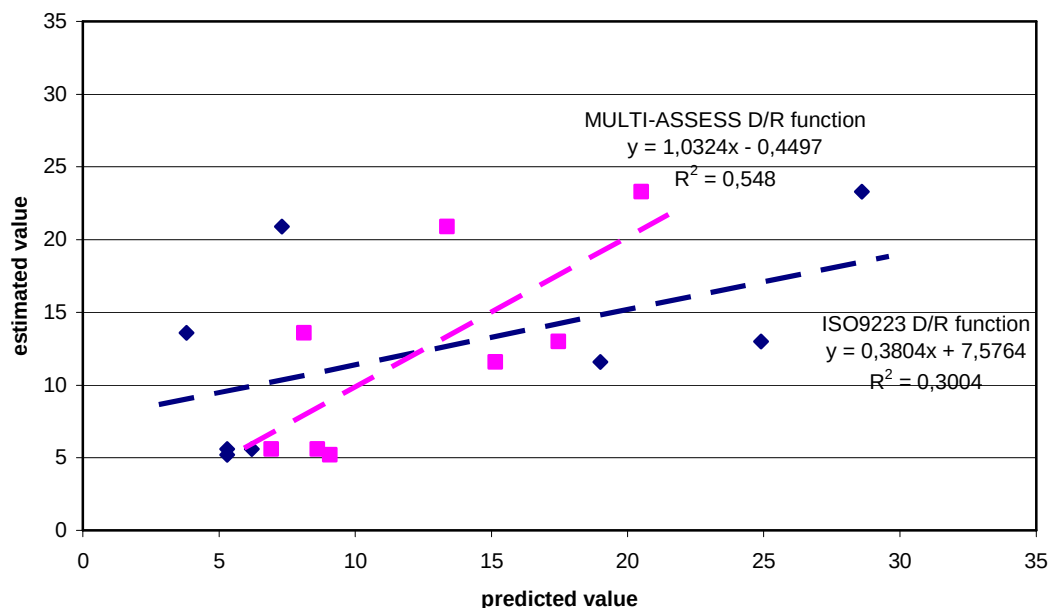
$$r_{corr} = 1.77 \cdot [SO_2]^{0.52} \cdot \exp(0.020 \cdot RH + f_{st}) + 0.102 \cdot [Cl]^{0.62} \cdot \exp(0.033 \cdot RH + 0.040 \cdot T) \quad [2]$$

In Table 9 the corrosion loss of carbon steel derived from real exposure are compared with corrosion loss values calculated from D/R functions. The estimated values can be calculated only for some test sites because the complete set of environmental data are still missing for the other test sites. The comparison of determined and estimated corrosion loss values are given in Figure 11, too. The MULTI-ASSESS equation shows the better correlation to estimated values of one-year corrosion loss than the ISO 9223 equation. The MULTI-ASSESS D/R function is based on the treatment of data set corresponding to environmental situation of exposure programme 2005/06.

Table 9 – Comparison of determined and estimated values of one-year corrosion loss of carbon steel ($\mu\text{m.a}^{-1}$)

Test site	determination	estimation – equation [1]	estimation – equation [2]
1 Prague	13,0	17,5	24,9
3 Kopisty	23,3	20,5	28,6
10 Bottrop	-		
13 Rome	13,1		
14 Casaccia	10,8		
15 Milan	14,6		
16 Venice	-		
21 Oslo	9,5		
23 Birkenes	13,6	8,1	3,8
24 Stockholm South	11,7		
26 Aspvreten	5,2		
27 Lincoln Cathedral	-	-	-
31 Madrid	5,6	6,9	5,3
33 Toledo	5,6	8,6	6,2
35 Lahemaa	11,3		
37 Dorset	11,0		
40 Paris	11,6	15,2	19,0
41 Berlin	28,1		
44 Svanvik	17,7		
45 Chaumont	5,2	9,1	5,3
50 Katowice	28,8		
51 Athens	15,4		
52 Riga	20,9	13,4	7,3

Figure 11 – The comparision of estimated and calculated values of one-year corrosion loss



3.4. Evaluation of other metal exposures (determined and estimated values)

On the base of previous exposure programs (MULTI-ASSESS – equation [3, 6], ISOCORRAG – equation [4, 7], UN ICP – equation [5, 8]) there were derived various D/R functions for zinc (equations 3 – 5) and for copper (equations 6 – 8):

$$ML_{Zn} = 1.82 + \{ 1.71 + 0.471[SO_2]^{0.22}e^{0.018Rh+f(T)} + 0.041Rain[H^+] + 1.37[HNO_3] \} t \quad [3]$$

$$r_{corr} = 0.0129 \cdot [SO_2]^{0.44} \cdot \exp(0.046 \cdot RH + f_{Zn}) + 0.0175 \cdot [Cl]^{0.57} \cdot \exp(0.008 \cdot RH + 0.085 \cdot T) \quad [4]$$

$$ML_{Zn} = 1.4[SO_2]^{0.22}e^{0.018Rh}e^{f(T)}t^{0.85} + 0.029Rain[H^+]t \quad [5]$$

$$ML_{Cu} = 3.12 + \{ 1.09 + 0.00201[SO_2]^{0.4}[O_3]Rh_{60} \cdot e^{f(T)} + 0.0878Rain[H^+] \} t \quad [6]$$

$$r_{corr} = 0.0053 \cdot [SO_2]^{0.26} \cdot \exp(0.059 \cdot RH + f_{Cu}) + 0.0125 \cdot [Cl]^{0.27} \cdot \exp(0.036 \cdot Rh + 0.049 \cdot T) \quad [7]$$

$$ML_{Cu} = 0.0027[SO_2]^{0.32}[O_3]^{0.79}Rh \cdot e^{f(T)}t^{0.78} + 0.050Rain[H^+]t^{0.89} \quad [8]$$

In Tables 10 and 11 the corrosion losses of zinc and copper derived from real exposure are compared with corrosion loss values calculated from D/R functions. For calculation based on D/R functions [4] and [7] the deposition of chlorides was used as background values S_0 ($3,0 \text{ mg} \cdot \text{m}^{-12} \cdot \text{d}^{-1}$). For zinc the determined and calculated values show very good correspondence mainly for test site with higher SO_2 pollution. For copper the data calculated from D/R function 6 are unusually higher – the parameter O_3 in this equation is used without any treatment and in location with relatively high concentration it affected the calculated values. It seems in real atmospheres the effect of O_3 on corrosion rate of copper is not as high as had been expected.

Table 10 – Comparison of determined and estimated values of corrosion loss of zinc ($\mu\text{m}\cdot\text{a}^{-1}$)

Test site	determination	estimation – equation [3]	estimation – equation [4]	estimation – equation [5]
Prague	0,44	1,06	1,10	1,21
Kopisty	1,23	1,07	1,38	1,31

Table 11 – Comparison of determined and estimated values of corrosion loss of copper ($\mu\text{m}\cdot\text{a}^{-1}$)

Test site	determination	estimation – equation [6]	estimation – equation [7]	estimation – equation [8]
Prague	0,31	2,05	0,76	0,73
Kopisty	1,39	2,29	0,85	1,07

3.5. Analysis of corrosion products of other metal exposed

Together with one-year exposure of carbon steel the zinc and copper samples had been exposed on the Czech test sites. For these samples the corrosion losses had been estimated by pickling. The detailed analysis of surface layers had been done by SEM/EDAX method (Figure 12 and Table 11).

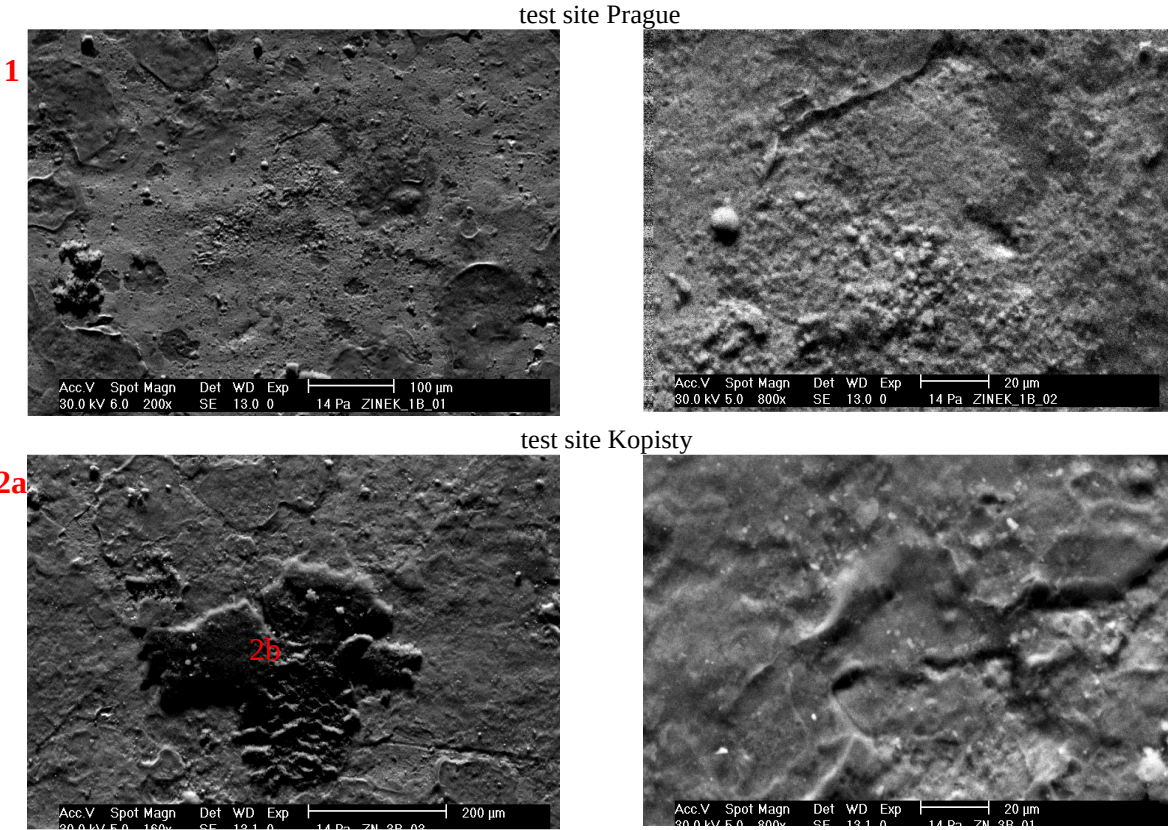
Table 11 – Results of SEM/EDAX analysis of metal surfaces

element	concentration (wt. %)					
	zinc			copper		
	analysed area					
	1	2a	2b	3	4a	4b
C	2,04	10,01	47,89	6,77	6,60	14,57
O	14,95	17,38	17,92	9,06	14,21	19,95
N	3,71	2,66	2,32	2,09	1,89	2,20
Al	1,05	1,24	0,88	0,62	2,25	1,70
Si	1,30	1,51	1,55	1,23	3,57	3,17
S	0,65	0,50	1,77	0,56	1,57	1,08
Cl	0,15	0,17	2,36	0,66	0,25	0,11
Cu	-	-	-	78,17	68,05	55,77
Zn	74.81	65.29	15.47	-	-	-

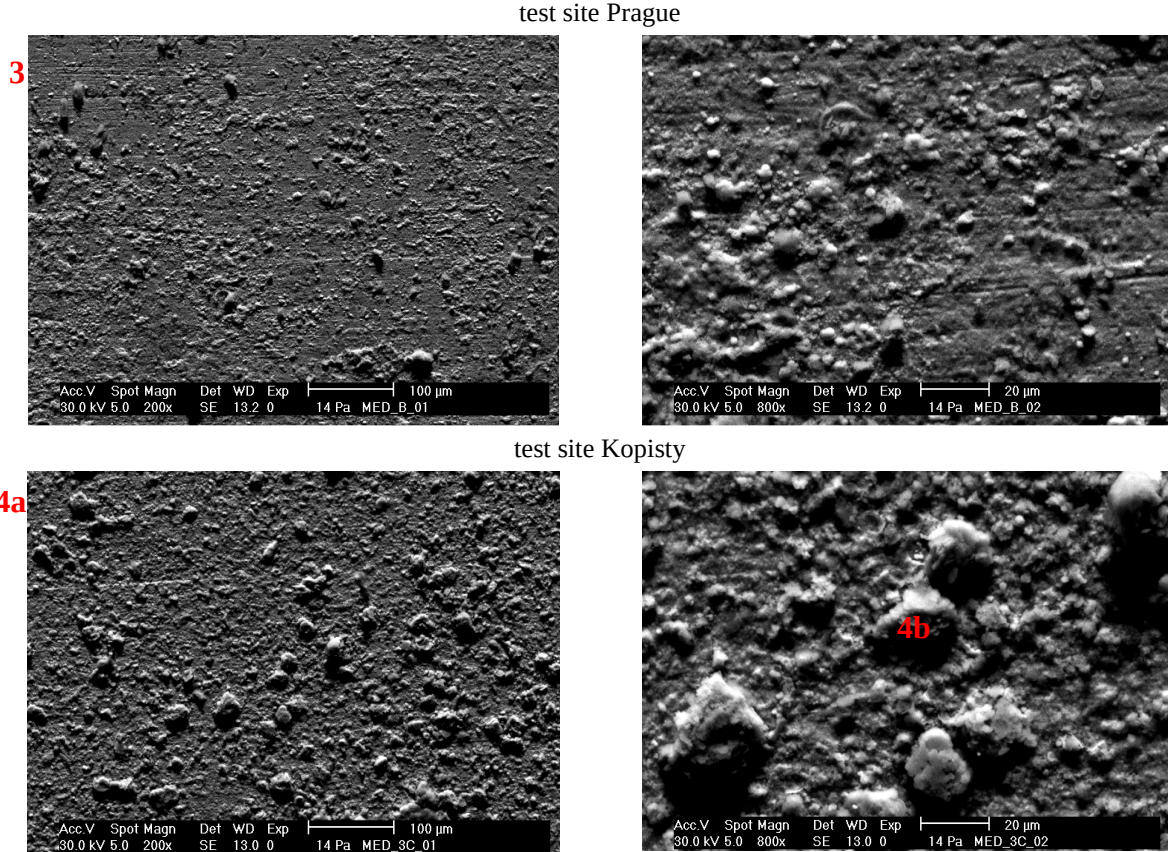
The surface layer at test site Kopisty shows higher amount of oxygen than the surface layer at Prague – the corrosion process is intensive. The surface layers from both test sites reflects the significant air pollution composition – at Prague it is nitrogen from higher NO_x and at Kopisty it is sulphur from higher SO_2 .

Figure 12 – The SEM/EDAX analysis of exposed samples' surface (200x and 800x magnitude)

zinc



copper



Conclusion

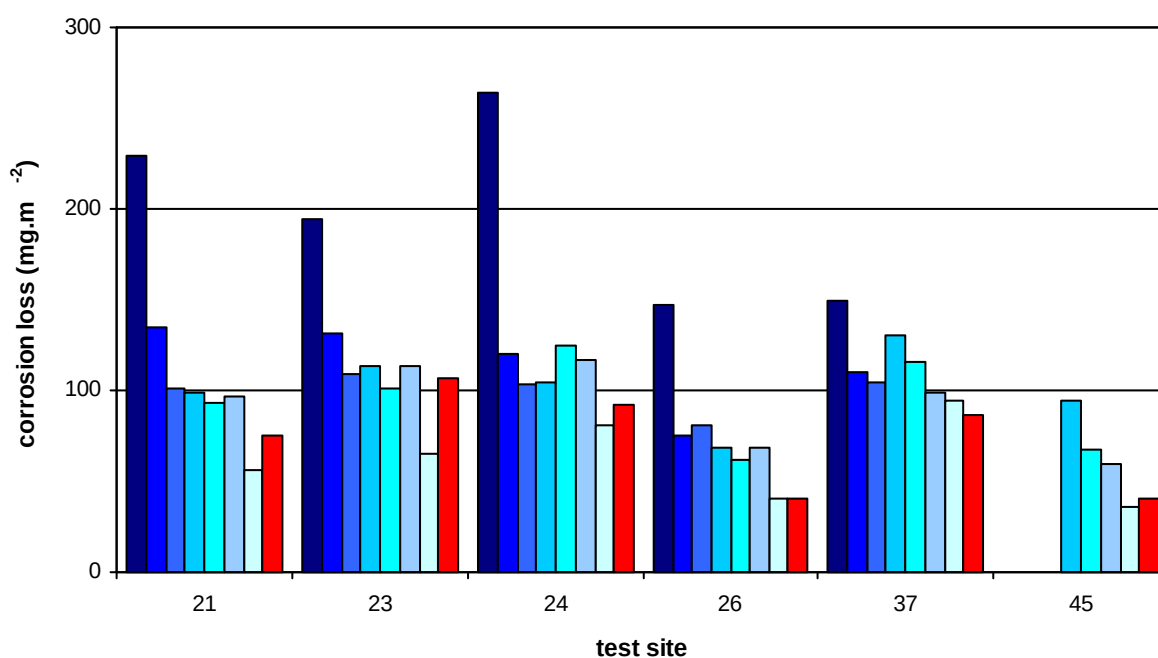
The systematic long-term evaluation of corrosion attack of carbon steel after 1 year exposure represents an important step in the ICP on materials activities. Repeated one-year exposures of carbon steel very clearly show the trend in decreasing of corrosivity of atmospheres and corrosion rate of carbon steel.

The evaluation presented in this report implicated:

- visual evaluation of exposed samples,
- evaluation of mass change of corroded samples,
- evaluation of corrosion losses after exposure,
- analysis of corrosion layers,
- comparison corrosion losses (determined, estimated).

During this exposure the corrosion loss of carbon steel increased in comparison to previous period significantly on test site No 23 (Birkenes) and partly on test sites No 21 and 24 (Figure 13). These data are necessary to be treated in context to corrosion loss of zinc and limestone which show out this phenomenon more significantly for these test sites.

Figure 13 – Corrosion loss (g.m^{-2}) after 1 year of exposure in various periods on selected test sites (since 1987/88 to 2005/06)



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- SCI - Swedish Corrosion Institute, Stockholm, Sweden
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- Ministry of the Environment of Estonia, Environmental Management and Technology, Tallinn, Estonia
- Ministry of the Environment, Canada
- LISA - Université Paris XII, Creteil, France
- EMPA - Swiss Federal Laboratories for Materials Testing and Research, Laboratory for Metallic Materials, Dübendorf, Switzerland

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Reference

1. Report No 51, Technical manual for the trend exposure programme 2005-2006, SCI, 2005
2. Report No 52 Environmental data report for the multi pollutant programme: November 2005 to October 2006, 2007